

Cultural Matching Analysis

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Overview

This analysis file examines the role of cultural matching and network opacity in predicting the persistence of social ties over time. We pool data across adjacent waves from the NetSense study to construct a person-period dataset, restricting the sample to non-kin, on-campus student ties. We estimate the probability of tie persistence as a function of cultural matching (closed-form domains and open-ended activities) and network opacity (the number of “Don’t Know” responses regarding an alter’s preferences) using mixed-effects logistic regression (`lme4::glmer`).

Main Effects Models

The following models assess the main effects of closed-form cultural matching, open-ended cultural matching, and network opacity on the odds of tie persistence. Model 1 shows that closed-form matches independently predict persistence. Model 2 introduces open-ended matches, demonstrating the value of incorporating both open and closed-form items. Model 3 incorporates network opacity (the number of “Don’t know” responses regarding an alter’s preferences). The main effects model reveals that a lack of shared cultural knowledge signals a weaker tie.

Marginal Effects

Figure Figure 1 visualizes the marginal effect (calculated as the change from the minimum to the maximum value) of closed-form matches, open-ended matches, and network opacity on the predicted probability of tie persistence.

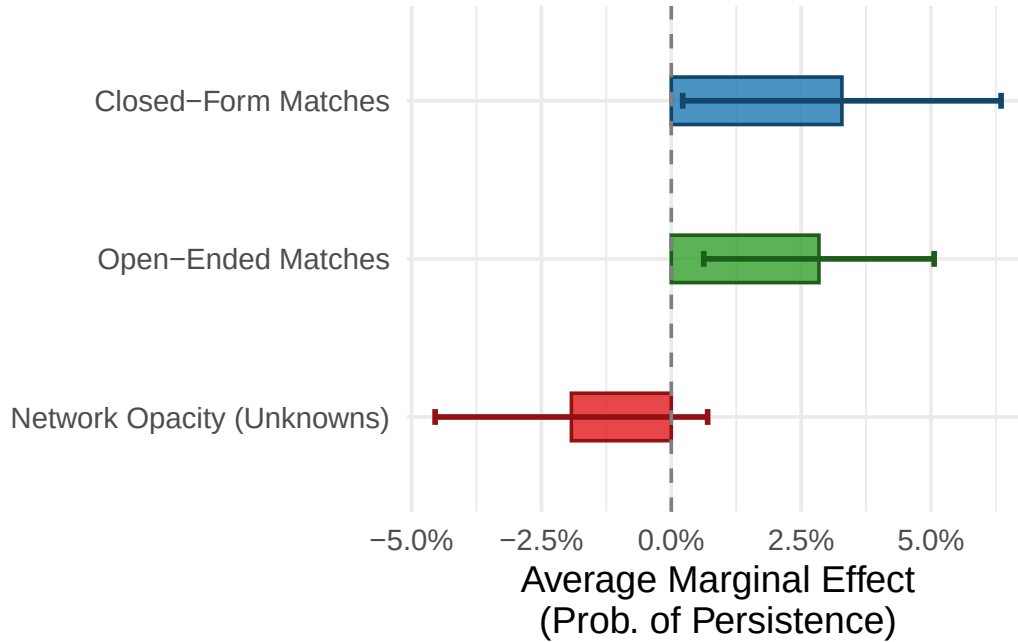


Figure 1: Marginal Effects of Cultural Matches and Network Opacity (Min to Max)

Subjective Closeness Interactions

To test whether the protective effects of cultural matching or the detrimental effects of network opacity depend on the strength of the tie, we estimated auxiliary interaction models between subjective closeness and each of the three cultural variables. While interaction terms in logistic regression models can be difficult to interpret directly from coefficients due to the non-linear link function, computing predicted probabilities across the range of the variables provides a clearer picture of the underlying dynamics.

Figure Figure 2 plots the marginal effects of the three cultural variables across the three levels of subjective closeness.

As the marginal effects plot demonstrates, the moderating role of subjective closeness varies by the type of cultural measure. For closed-form matches and network opacity, there is a strong interaction with tie strength: the marginal effects are steepest for “Not Close” ties, indicating

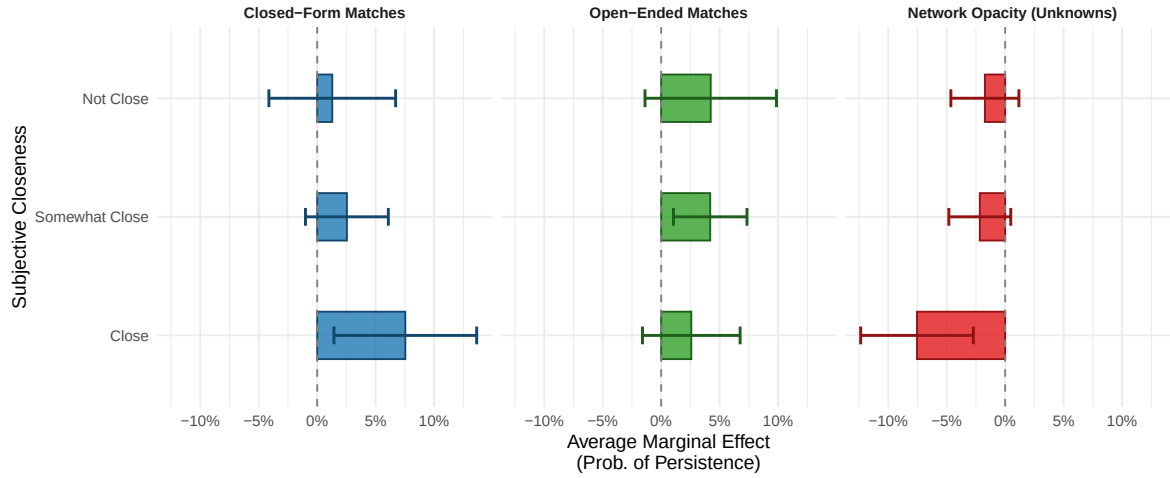


Figure 2: Marginal Effects of Cultural Matching and Network Opacity by Subjective Closeness

that broad cultural alignment and mutual cultural knowledge provide critical scaffolding that helps weak ties survive. Conversely, for ties that are already “Close,” the slopes for closed-form matches and opacity are much flatter, suggesting these strong ties are more resilient to decay regardless of these factors. In contrast, open-ended matches do not show as strong of an interaction; their protective effect remains relatively consistent across different levels of subjective closeness.

Appendix: Descriptive Statistics

Robustness Check: Ego Fixed Effects

To ensure the robustness of our main findings, we estimated an **Ego Fixed-Effects** model using conditional logistic regression (`survival::clogit`). This stringent test completely controls for all unobserved, time-invariant ego characteristics by only leveraging within-ego variance (i.e., comparing alters retained vs. dropped by the same ego).